

EXAMPLE 12.16

A 2 kW load takes a current of 16 A from a 200 V a.c. supply. Calculate the active and reactive components of the current.

SOLUTION

$$P = 2 \text{ kW} = 2 \times 1000 \text{ W}$$

$$I = 16 \text{ A}, V = 200 \text{ volts}$$

$$P = VI \cos \phi$$

$$\cos \phi = \frac{P}{VI}$$

$$= \frac{2000}{200 \times 16} = 0.625$$

$$\phi = 51.32^\circ, \sin \phi = 0.78$$

Active component of current

$$I_a = I \cos \phi$$

$$= 16 \times 0.625 = 10 \text{ A}$$

Reactive component of current

$$I_r = I \sin \phi$$

$$= 16 \times 0.78 = 12.48 \text{ A}$$

EXAMPLE 12.17

A single-phase load of 30 kW at 0.6 power factor lagging is fed from 200 V a.c. supply. Calculate the kVA and kVAR of the load.

SOLUTION

$$P = 30 \text{ kW}, \cos \phi = 0.6$$

$$\therefore \sin \phi = 0.8$$

$$\text{kVA} \cos \phi = \text{kW}$$

$$\text{kVA} = \frac{\text{kW}}{\cos \phi}$$

$$= \frac{30}{0.6} = 50$$

$$\text{kVA } r = \text{kVA} \sin \phi$$

$$= 50 \times 0.8 = 40$$

EXAMPLE 16.3

Show that the power consumed by three identical phase loads connected in delta is equal to three times the power consumed when the phase loads are connected in star.

SOLUTION

Let

 V_L = line voltage R = resistance of each phase load Z = impedance of each phase load*Star connection*

$$V_P = V_L / \sqrt{3}$$

$$P_Y = 3 V_P I_P \cos \phi = 3 V_P \cdot \frac{V_P}{Z} \cdot \frac{R}{Z} = 3 V_P^2 \frac{R}{Z^2}$$

$$= 3 \left(\frac{V_L}{\sqrt{3}} \right)^2 \frac{R}{Z^2} = V_L^2 \frac{R}{Z^2}$$

Delta connection

$$V_P = V_L$$

$$P_\Delta = 3 V_P I_P \cos \phi$$

$$= 3 V_P \cdot \frac{V_P}{Z} \cdot \frac{R}{Z} = 3 V_P^2 \frac{R}{Z^2} = 3 V_L^2 \frac{R}{Z^2}$$

By comparison, $P_\Delta = 3 P_Y$ **EXAMPLE 16.4**

Three identical resistors of 20Ω are connected in star to a 415 V, three-phase, 50 Hz supply. (a) Calculate the total power taken by the load. (b) Also calculate the power consumed in the resistors if they are connected in delta to the same supply. (c) If one of the resistors is open-circuited in each case, calculate the power consumed.

SOLUTION*(a) Star connection*

$$V_P = V_L / \sqrt{3} = 415 / \sqrt{3} = 240 \text{ V}$$

$$I_P = \frac{V_P}{R} = \frac{240}{20} = 12 \text{ A}$$

$$P = 3 I_P^2 R = 3 \times 12^2 \times 20 = 8640 \text{ W}$$

(b) Delta connection

$$V_P = V_L = 415 \text{ V}$$

$$I_P = \frac{V_P}{R} = \frac{415}{20} = 20.75 \text{ A}$$

$$P = 3 I_P^2 R = 3 (20.75)^2 \times 20 = 25833 \text{ W}$$

(c) One resistor open-circuited in star connection

From Fig. 16.18

$$\text{current } I_1 = \frac{V_L}{2R} = \frac{415}{2 \times 20} = 10.375 \text{ A}$$

Power consumed

$$P = 2 I_1^2 R = 2 (10.375)^2 \times 20 = 4305 \text{ W}$$

(d) One resistor open-circuited in delta connection

This arrangement is shown in Fig. 16.19.

Power consumed

$$P = \frac{V_L^2}{R} + \frac{V_L^2}{R} = 2 \frac{V_L^2}{R} = 2 \times \frac{(415)^2}{20} = 17222 \text{ W}$$

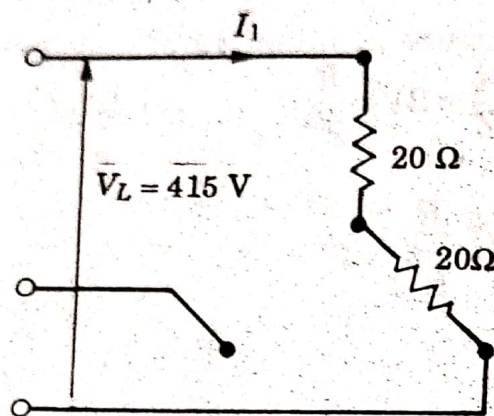


Fig. 16.18

EXAMPLE 16.7

A three-phase load consists of three similar inductive coils, each of resistance $50\ \Omega$ and inductance $0.3\ \text{H}$. The supply is $415\ \text{V}$, $50\ \text{Hz}$. Calculate : (a) the line current; (b) the power factor; and (c) the total power when the load is (i) star-connected and (ii) delta-connected.

SOLUTION

(i) Star connection

The star-connected load is shown in Fig. 15.20a. One phase of the load is shown in Fig. 15.20b.

Inductive reactance per phase

$$X_{LP} = 2\pi fL = 2\pi \times 50 \times 0.3 = 94.2\ \Omega$$

Resistance per phase $R_P = 50\ \Omega$

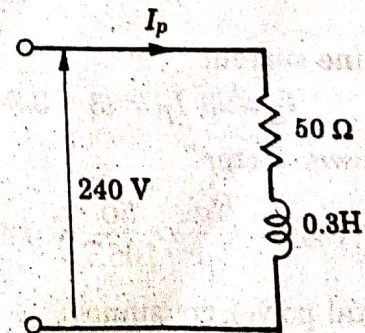
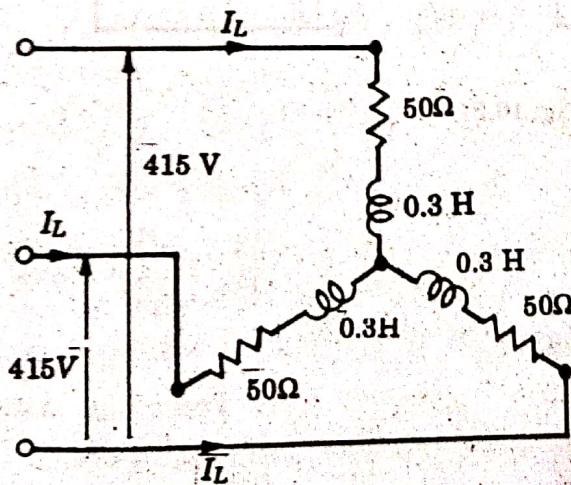


Fig. 16.20

Impedance per phase

$$Z_P = \sqrt{R^2 + X_{LP}^2} = \sqrt{(50)^2 + (94.2)^2} = 106.7 \Omega$$

$$V_P = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 240 \text{ V}$$

$$I_P = \frac{V_P}{Z_P} = \frac{240}{106.7} = 2.25 \text{ A}$$

(a) Line current

$$I_L = I_P = 2.25 \text{ A}$$

(b) Power factor

$$\cos \phi = \frac{R_P}{Z_P} = \frac{50}{106.7} = 0.469 \text{ (lagging)}$$

(c) Total power consumed

$$P = 3V_P I_P \cos \phi = 3 \times 240 \times 2.25 \times 0.469 = 760 \text{ W}$$

Alternatively

$$P = 3 I_P^2 R_P = 3(2.25)^2 \times 50 = 759.4 \text{ W}$$

(ii) Delta connection

The delta-connected load is shown in Fig. 16.21a. One phase of the load is shown in Fig. 16.21b.

$$V_P = V_L = 415 \text{ V}$$

$$I_P = \frac{V_P}{Z_P} = \frac{415}{106.7} = 3.9 \text{ A}$$

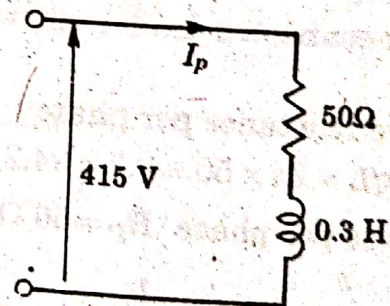
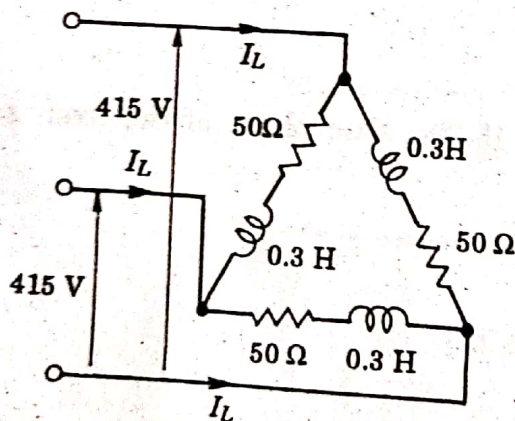


Fig. 16.21

(a) Line current

$$I_L = \sqrt{3} I_P = \sqrt{3} \times 3.9 = 6.75 \text{ A}$$

(b) Power factor

$$\cos \phi = \frac{R_P}{Z_P} = \frac{50}{106.7} = 0.469 \text{ (lagging)}$$

(c) Total power consumed

$$P = 3V_P I_P \cos \phi = 3 \times 415 \times 3.9 \times 0.469 = 2277 \text{ W}$$

$$\text{Alternatively, } P = 3 I_P^2 R_P = 3 \times (3.9)^2 \times 50 = 2281 \text{ W}$$

EXAMPLE 16.11

A balanced delta-connected load of $60 \angle 30^\circ \Omega$ per phase is connected across a 3-phase, 400 V, 50 Hz supply. Calculate the line current, power factor and power consumed.

SOLUTION

$$Z_p = 60 \angle 30^\circ$$

$$V_p = V_L = 400 \text{ V}$$

Let V_p be taken as reference phasor.

$$\therefore V_p = 400 \angle 0^\circ \text{ V}$$

By Ohm's law

$$I_p = \frac{V_p}{Z_p} = \frac{400 \angle 0^\circ}{60 \angle 30^\circ} = 6.66 \angle -30^\circ \text{ A}$$

$$\text{Line current } I_L = \sqrt{3} I_p = \sqrt{3} \times 6.66 = 11.53 \text{ A}$$

$$\text{p.f.} = \cos \phi = \cos (-30^\circ) = 0.866 \text{ (lagging)}$$

Total 3-phase power

$$P_{3\phi} = 3 V_p I_p \cos \phi = 3 \times 400 \times \frac{400}{60} \times 0.866 = 6928 \text{ W}$$

EXAMPLE 16.12

A balanced 3-phase star-connected load of 120 kW takes a leading current of 100 A when connected across a 3-phase, 3.3 kV, 50 Hz supply. Determine the impedance, resistance, capacitance and power factor of the load.

SOLUTION

$$V_L = 3.3 \text{ kV} = 3.3 \times 10^3 = 3300 \text{ V}$$

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{3300}{\sqrt{3}} = 1905.31 \text{ V}$$

$$I_L = I_p = 100 \text{ A}, f = 50 \text{ Hz}$$

$$P_{3\phi} = \sqrt{3} V_L I_L \cos \phi$$

$$120 \times 10^3 = \sqrt{3} \times 3300 \times 100 \times \cos \phi$$

$$\therefore \cos \phi = 0.2099, \phi = 77.88^\circ$$

By Ohm's Law

$$Z_p = \frac{V_p}{I_p} = \frac{1905.31 \angle 0^\circ}{100 \angle 77.88^\circ} = 19.05 \angle -77.88^\circ = 3.99 - j 18.62$$

$$\text{Also } Z_p = R_p - jX_{cp}$$

$$\therefore R_p = 3.99 \, \Omega, X_{cp} = 18.62 \, \Omega$$

$$X_{cp} = \frac{1}{2\pi f C_p}$$

$$C_p = \frac{1}{2\pi \times 50 \times 18.62} = 179.95 \times 10^{-6} \, \text{F} = 179.95 \, \mu\text{F}$$